§6 Minimal structure of ACS relations — part 3 of 4 (sign classification of Q, distinction from environment coupling, finite-window aggregation, relation to T-STAB)

Source: `paper/section6_relational.tex`, lines 1332–1791 (sign classification of Q; non-reference between Q and the structural environment coupling condition; environment coupling with negative relation quality; the relation of Q to feasibility change; division of roles with temporal aggregation; finite-window aggregation of Q; relation to T-STAB), based on commit e6e12cd, with the author's correction to `rem:Q_connection` recorded in ledger entry L-A3.195. Status: draft. Part 2 is `06_relational_2.md`; part 4 (`06_relational_4.md`, 1792–2171: representative categories and six families, scope, hand-off, summary) follows. This part defines no assumption; the three assumptions of §6 are in parts 1 and 2.

Sign classification of Q

Definition (`def:Q_sign`): classification of relation quality by the sign of Q. For an evaluable relation, that is, at time t with $Y_j \in \mathcal{D}_{Q,i,t}$, the sign of $Q_{i\leftarrow j,t}$ is used for description as follows: $Q_{i\leftarrow j,t} > 0$ is called *positive-side relation quality*, $Q_{i\leftarrow j,t} = 0$ *neutral relation quality*, and $Q_{i\leftarrow j,t} < 0$ *negative-side relation quality*. The classification is a descriptive classification of the direction of structural evaluation relating to the maintenance of feasibility of the ACS concerned. It does not judge the ethical or normative good or bad of the external object or the relation.

$Q = 0$ is the neutral boundary of a defined evaluation (`rem:Q_threshold`). In `def:Q_sign`, $Q = 0$ is not set by introducing a separate exogenous universal threshold $Q_{\text{threshold}}$; it is the neutral boundary of the signed component representation of `def:Q`. Accordingly $Q > 0$, $Q = 0$ and $Q < 0$ express the positive side, neutrality and the negative side of this quality evaluation. The statement $Q = 0$ is meaningful only when $Y_j \in \mathcal{D}_{Q,i,t}$. Not being evaluable, $Y_j \notin \mathcal{D}_{Q,i,t}$, must not be identified with $Q = 0$ (compare `rem:Q_undefined`).

Distinction of Q from structural environment coupling and realized external interaction

Proposition (`prop:Q_not_RO_condition`): mutual non-reference of Q and the structural environment coupling condition at the level of the defining formulas. (i) Whether the structural environment coupling condition $\mathrm{EC}(x_t)$ (`def:env_coupling`) holds is determined by x_t, f, U, E_{ext} and π_E alone; its definition contains neither the evaluable domain $\mathcal{D}_{Q,i,t}$ nor the value of Q. (ii) The relation quality evaluation $Q_{i\leftarrow j,t}$ (`def:Q`) is determined for an evaluable relation with $Y_j \in \mathcal{D}_{Q,i,t}$, and its definition does not contain $\mathrm{EC}(x_t)$. The proof is by inspection of the definitions. For (i), the condition of `def:env_coupling` consists only of quantification over $u \in U$ and $e, e' \in E_{\text{ext}}$ and the inequality $\pi_E(f(x_t, u, e)) \neq \pi_E(f(x_t, u, e'))$. For (ii), `def:Q_rule`, `def:Q_domain` and `def:Q` contain no occurrence of `def:env_coupling`.

The proposition states only that the two definitions do not contain each other; it does not show that no implication holds between them. This paper does not state any of the following as a general consequence: from $Q_{i\leftarrow j,t} > 0$ or $Q_{i\leftarrow j,t} < 0$ for an evaluable relation, whether $\mathrm{EC}(x_t)$ holds; from $\mathrm{EC}(x_t)$, for a particular Y_j, either $Y_j \in \mathcal{D}_{Q,i,t}$ or $Q_{i\leftarrow j,t} > 0$; and from the sign of Q, the truth or falsity of EU_t.

Q is not identified with the realized external interaction (`rem:Q_chi_distinction`). The realized external interaction indicator $\chi_{\mathrm{EI},t}$ of §3 (`subsec:env_coupling`) is also not the same as Q. $\chi_{\mathrm{EI},t} = 1$ means that a boundary-crossing external interaction is realized at time t; this paper does not state from it alone, for a particular Y_j, either $Y_j \in \mathcal{D}_{Q,i,t}$ or $Q_{i\leftarrow j,t} > 0$ as a general consequence. Conversely, Phase 1 does not exclude that a relation quality evaluation is defined on the basis of past interactions, the internal model, memory or other relational information; hence $\chi_{\mathrm{EI},t} = 1$ is not stated as a general consequence of $Y_j \in \mathcal{D}_{Q,i,t}$ alone. $\mathrm{EC}(x_t)$, $\chi_{\mathrm{EI},t}$, EU_t and Q are therefore treated as mutually distinct.

Environment coupling with negative relation quality

Definition (`def:contaminated_ro`): environment coupling with negative relation quality (classification C). At time t, if the structural environment coupling condition $\mathrm{EC}(x_t)$ holds and, for some evaluable external object $Y_j \in \mathcal{D}_{Q,i,t}$, $Q_{i\leftarrow j,t} < 0$, the configuration is called, with respect to that relation Y_j, *environment coupling with negative relation quality*. That is, it is the descriptive configuration $\mathrm{EC}(x_t) \wedge \exists\, Y_j \in \mathcal{D}_{Q,i,t} : Q_{i\leftarrow j,t} < 0$, and it does not judge the good or bad of the external object or the relation.

Scope (`rem:contaminated_ro_scope`). The notion is a descriptive concept for distinguishing the structural configuration in which, although the structural environment coupling condition holds, at least one evaluable relation receives a negative-side quality evaluation. It does not mean that structural environment coupling is itself "bad", that relations with the outside should be closed, or that the external object is ethically harmful. This paper also does not state from $Q_{i\leftarrow j,t} < 0$ alone any of $\Pi(x_{t+1}) < \Pi(x_t)$, $A_t = \emptyset$ or $\mathrm{Col}_{i,t} = 1$ as a general consequence. Q expresses a direction of relation quality evaluation, but a quantitative connection between that evaluation and the actual state transition involves f, the realized action, the realized external environment state, the internal state and, where needed, additional conditions.

Negative-side relation quality is not the same as environment coupling with negative relation quality (`rem:contaminated_status`). Negative-side relation quality $Q_{i\leftarrow j,t} < 0$ is a description of one evaluable relation. Environment coupling with negative relation quality additionally requires that $\mathrm{EC}(x_t)$ holds. The two are therefore not the same concept. Further, environment coupling with negative relation quality is not a fourth principle added to the three principles, not a separate form of the structural environment coupling condition, not a separate kind of condition on the time-indexed satisfaction predicate of EU, and not a new failure state.

Relation of Q to feasibility change

The connection between the sign of Q and $\Delta\Pi$ is not stated as a general law (`rem:Q_connection`). Even when, for $t \geq 1$ at which both states x_{t-1} and x_t are defined, the time difference $\Delta\Pi_t := \Pi(x_t) - \Pi(x_{t-1})$ is considered, this paper does not state either of the implications $Q_{i\leftarrow j,t} < 0 \Rightarrow \Delta\Pi_t < 0$ and $Q_{i\leftarrow j,t} > 0 \Rightarrow \Delta\Pi_t > 0$ as a general consequence. This does not establish that either implication fails. The stated reason is that Q is an overall relation quality evaluation about a particular relation under evaluation, whereas $\Delta\Pi_t$ is determined as the result of the whole realized state transition of the ACS. No general necessary-and-sufficient relation is therefore placed between the sign of Q and the direction of change of Π. To connect the two quantitatively for a particular ACS or relation type, an additional structural assumption or dynamical model is needed.

Division of roles with temporal aggregation

Temporal aggregation of Q is treated in later subsections (`rem:Q_temporal_deferred`). Q in this subsection defines the quality evaluation of an individual evaluable relation at time t. The finite-window average is treated in the latter half of the subsection, and the time integration of principle-related evaluation (T-STAB) in `subsec:s3_tstab`. This subsection therefore does not substitute the conventional value 0 into a time average when no evaluable Q exists, does not fix a single relation under evaluation as a universal "representative relation", and does not convert Q directly into the time-indexed satisfaction predicate of EU or into a degree of the structural environment coupling condition.

Finite-window aggregation of the relation quality Q

Definition (`def:Q_window_average`): finite-window evaluation of relation quality. Fix a finite window width $W_Q \in \mathbb{N}$ with $W_Q \geq 1$. The finite window for Q at time $t \in \mathbb{N}_0$ is $\mathcal{W}_t^Q := \{r \in \mathbb{N}_0 \mid \max(0, t - W_Q + 1) \leq r \leq t\}$. For an external object $Y_j \in \mathcal{Y}_{-i}$, let $\mathbb{T}_{i\leftarrow j,t}^Q := \{r \in \mathcal{W}_t^Q \mid Y_j \in \mathcal{D}_{Q,i,r}\}$ be the set of times in this window at which $Q_{i\leftarrow j,r}$

is defined. Only when $\mathbb{T}^{Q}_{i\leftarrow j,t} \neq \emptyset$, the finite-window average of the relation is defined as $\bar{Q}_{i\leftarrow j,W_Q,t} := \frac{1}{|\mathbb{T}^{Q}_{i\leftarrow j,t}|} \sum_{r\in\mathbb{T}^{Q}_{i\leftarrow j,t}} Q_{i\leftarrow j,r} \in [-1,1]$. When $\mathbb{T}^{Q}_{i\leftarrow j,t} = \emptyset$, $\bar{Q}_{i\leftarrow j,W_Q,t}$ is *undefined*.

No conventional value 0 **for a window without evaluation of** Q **(`rem:Q_window_undefined`).** `def:Q_window_average` keeps the distinction "not evaluable ≠ neutral quality" after temporal aggregation. When no evaluable Q exists in the finite window, $\bar{Q}_{i\leftarrow j,W_Q,t} := 0$ is not set. Phase 1 also places no universal rule that aggregates $\bar{Q}_{i\leftarrow j,W_Q,t}$ over several external objects into a single overall $\bar{Q}_{i,t}$. Which relation is given priority as the object of evaluation, by which operator several relations are integrated, and how object-dependent weights are updated can depend on the implementation and on later dynamics (within the present framework).

Temporal aggregation of Q **does not redefine EU or structural environment coupling (`rem:Q_window_not_RO`).** The signs $\bar{Q}_{i\leftarrow j,W_Q,t} > 0$, $= 0$ and < 0 are time-aggregated quality descriptions of one evaluable relation. This paper does not state from these signs, as a general consequence, the truth or falsity of EU_r at times in the window, or whether the structural environment coupling condition $\mathrm{EC}(x_r)$ (`def:env_coupling`) holds. The finite-window evaluation of Q is therefore neither a condition of EU nor a structural environment coupling condition; it is treated as an aggregation of relation quality in the time direction.

Relation to T-STAB

Q **is not built into the universal condition of T-STAB (`rem:Q_not_tstab_core`).** This paper does not build into T-STAB (`def:tstab`) a lower bound on relation quality such as $\bar{Q}_{i\leftarrow j,W_Q,t} \geq Q^{\min}$ as a universal required condition. The stated reasons are that Q is a quantity relative to a particular evaluable external relation, so the evaluable domain $\mathcal{D}_{Q,i,t}$ can be empty depending on time or object, and that Phase 1 sets no universal rule aggregating several relations under evaluation into a single overall quality. The temporal aggregation of Q is therefore kept as a channel for temporal evaluation of relation quality that runs in parallel to T-STAB. This paper does not exclude building some aggregate of a relation or of several relations into an internal reference condition of T-STAB type in a particular ACS implementation; in that case the choice of object, the aggregation operator, the threshold and the treatment of missing evaluation must be defined explicitly in addition.

Temporal evaluation of environment coupling with negative relation quality (`rem:contaminated_ro_temporal`). Environment coupling with negative relation quality is described, as a configuration at each time, by $\mathrm{EC}(x_t) \wedge \exists Y_j \in \mathcal{D}_{Q,i,t} : Q_{i\leftarrow j,t} < 0$ (`def:contaminated_ro`). When $\bar{Q}_{i\leftarrow j,W_Q,t}$ is defined for a particular Y_j by `def:Q_window_average`, its sign can be used in the description. However, a negative window average does not imply that $Q_{i\leftarrow j,r}$ is defined and negative at each time in the window. To state temporal persistence of negative-side relation quality, the definedness and the sign at each time in the window are to be checked separately. Further, this paper does not state from $\bar{Q}_{i\leftarrow j,W_Q,t} < 0$, as a general consequence, that some time-indexed satisfaction predicate is false at a time in the window, that the structural environment coupling condition fails to hold, or the non-satisfaction of the T-STAB condition, blockage or collapse. The finite-window average is an aggregation of relation quality evaluation in the time direction and is not itself the result of a state transition.

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